

Superposition Principle of Homogeneous Equations:

For a homogeneous n -th order ODE of the form:

$$a_n(x) \cdot \frac{d^n y}{dx^n} + a_{n-1}(x) \cdot \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \cdot \frac{dy}{dx} + a_0(x) y = 0$$

Suppose its solutions are $y_1, y_2, \dots, y_n$.

↳ Then the linear combinations of these solutions:

$$y^* = C_1 \cdot y_1 + C_2 \cdot y_2 + \dots + C_n \cdot y_n$$

are also solutions of the DE.

↖ $C_1, C_2, \dots, C_n$
are arbitrary
constants

Linearly Independent functions:

Two functions $f(x)$ and $g(x)$ are said to be linearly independent if neither function is a constant multiple of the other.

$$\text{i.e. } f(x) \neq C_1 \cdot g(x) \quad \text{and} \quad g(x) \neq C_2 \cdot f(x)$$

Solving Homogeneous Linear Eqs with constant coefficients:

↳ For: 1st or higher order ODE + Homogeneous + Linear

↳ The DE is of the form:

$$ay'' + by' + cy = 0$$

← a, b, c are
constants

↳ The solution to this DE can be one of 3 different forms. To determine which one, we use the auxiliary of the DE.

↳ Let $y = e^{mx}$ be a trial solution to the DE.

↳ Substituting $y = e^{mx}$ into the DE, we get:

$$a \cdot m^2 \cdot e^{mx} + b \cdot m \cdot e^{mx} + c \cdot e^{mx} = 0$$

$$\Rightarrow am^2 + bm + c = 0 \quad \leftarrow \text{The auxiliary eqn}$$

↳ We solve this eqn for m , then of the DE

depending on the roots of this equation,

the solution can take on different forms:

<u>Discriminant</u>	<u>Roots</u>	<u>Form of the solution</u>
$b^2 - 4ac > 0$	Real & Unequal roots m_1 and m_2	$y = C_1 \cdot e^{m_1 x} + C_2 \cdot e^{m_2 x}$
$b^2 - 4ac = 0$	Real & Equal / Repeated roots i.e. $m_1 = m_2$	$y = C_1 \cdot e^{m_1 x} + C_2 x \cdot e^{m_1 x}$
$b^2 - 4ac < 0$	Imaginary / Complex Roots i.e. $m_1 = a + ib$ $m_2 = a - ib$	$y = e^{ax} (C_1 \cdot \cos bx + C_2 \cdot \sin bx)$

↳ The solution obtained in this way is also a general solution.

* The roots are complex conjugates of each other.

(e.g.) Solve the following DEs:

a) $2y'' - 5y' - 3y = 0$

b) $y'' - 10y' + 25y = 0$

c) $4y'' + 4y' + 17y = 0$; $y(0) = -1$, $y'(0) = 2$

Soln a) $2y'' - 5y' - 3y = 0$

Let $y = e^{mx}$ $\therefore y' = me^{mx}$ and $y'' = m^2 e^{mx}$

Substituting these into the DE:

$$2m^2 e^{mx} - 5m e^{mx} - 3e^{mx} = 0$$

$$\Rightarrow 2m^2 - 5m - 3 = 0$$

$$\Rightarrow m = \frac{-(-5) \pm \sqrt{(-5)^2 - 4 \cdot 2 \cdot (-3)}}{2 \cdot 2}$$

$$= \frac{5 \pm \sqrt{25 + 24}}{4}$$

$$= \frac{5 \pm 7}{4} = \frac{12}{4} \quad \text{or} \quad \frac{-2}{4}$$

$$\Rightarrow m = 3 \quad \text{or} \quad m = -\frac{1}{2} \quad \leftarrow \text{So roots are distinct}$$

$\therefore$ General Solution is of the form:

$$y = C_1 \cdot e^{m_1 x} + C_2 \cdot e^{m_2 x}$$

$$\Rightarrow y = C_1 \cdot e^{3x} + C_2 \cdot e^{-\frac{x}{2}}$$

b) $y'' - 10y' + 25y = 0$

Let $y = e^{mx}$ $\therefore y' = me^{mx}$ and $y'' = m^2 e^{mx}$

Substituting these into the DE:

$$m^2 e^{mx} - 10m \cdot e^{mx} + 25 e^{mx} = 0$$

$$\Rightarrow m^2 - 10m + 25 = 0$$

$$\Rightarrow m^2 - 5m - 5m + 25 = 0$$

$$\Rightarrow m(m - 5) - 5(m - 5) = 0 \quad \Rightarrow (m - 5)^2 = 0$$

$$\Rightarrow m = 5 \quad | \quad \Rightarrow m = 5 \quad \leftarrow \text{So roots repeat.}$$

$\therefore$ General solution is of the form:

$$y = C_1 e^{5x} + C_2 x e^{5x}$$

c) $4y'' + 4y' + 17y = 0$; $y(0) = -1$, $y'(0) = 2$

Let $y = e^{mx}$ $\therefore y' = m e^{mx}$ and $y'' = m^2 e^{mx}$

Substituting these into the DE:

$$4m^2 + 4m + 17 = 0$$

$$\Rightarrow m = \frac{-4 \pm \sqrt{4^2 - 4 \cdot (4) \cdot (17)}}{2 \cdot 4}$$

$$= \frac{-4 \pm \sqrt{-256}}{8} = \frac{-4 \pm 16\sqrt{-1}}{8}$$

$$= \frac{-1 \pm 4i}{2} \rightarrow \therefore \left. \begin{array}{l} m_1 = -\frac{1}{2} + 2i \\ m_2 = -\frac{1}{2} - 2i \end{array} \right\} \text{Here, } a = -\frac{1}{2} \\ b = 2$$

$\therefore$ The solution to the DE is:

$$y = e^{-\frac{x}{2}} (C_1 \cos 2x + C_2 \sin 2x)$$

$\rightarrow$ Given $y(0) = -1$:

$$-1 = e^0 (C_1 \cos 0 + C_2 \sin 0)$$

$$\Rightarrow -1 = 1 (C_1 \cdot 1 + C_2 \cdot 0)$$

$$\Rightarrow -1 = C_1 + 0 \Rightarrow C_1 = -1$$

$\rightarrow$ Now, given $y'(0) = 2$:

$$y_1 = -\frac{1}{2} e^{-\frac{x}{2}} (C_1 \cos 2x + C_2 \sin 2x) + e^{-\frac{x}{2}} (-2C_1 \sin 2x + 2C_2 \cos 2x)$$

$$\Rightarrow 2 = -\frac{1}{2} \cdot 1 (C_1 \cdot 1 + C_2 \cdot 0) + 1 (-2C_1 \cdot 0 + 2C_2 \cdot 1)$$

$$\Rightarrow 2 = -\frac{1}{2} \cdot C_1 + 2C_2 \quad \Rightarrow 2 = \frac{1}{2} + 2C_2 \quad \Rightarrow C_2 = \frac{3}{4}$$

$$\text{Hence, } y = e^{-x/2} \left(-\cos 2x + \frac{3}{4} \sin 2x \right)$$

(e.g.) Solve $16y^{(4)} + 24y'' + 9y = 0$

Solⁿ Let $y = e^{mx}$

$\therefore$ The DE becomes:

$$16m^4 \cdot e^{mx} + 24m^2 \cdot e^{mx} + 9e^{mx} = 0$$

$$\Rightarrow 16m^4 + 24m^2 + 9 = 0$$

$$\Rightarrow 16m^4 + 12m^2 + 12m^2 + 9 = 0$$

$$\Rightarrow 4m^2(4m^2 + 3) + 3(4m^2 + 3) = 0$$

$$\Rightarrow (4m^2 + 3)^2 = 0$$

$$\Rightarrow 4m^2 + 3 = 0$$

$$\Rightarrow m = \pm \sqrt{\frac{-3}{4}}$$

$$\Rightarrow m = \pm i \sqrt{\frac{3}{4}}$$

$$\Rightarrow 4m^2 + 3 = 0$$

$$\Rightarrow m = \pm \sqrt{\frac{-3}{4}}$$

$$\Rightarrow m = \pm i \sqrt{\frac{3}{4}}$$

$$\therefore y = e^{0x} \left(C_1 \cdot \cos\left(x\sqrt{\frac{3}{4}}\right) + C_2 \cdot \sin\left(x\sqrt{\frac{3}{4}}\right) \right)$$

$$+ e^{0x} \cdot x \left(C_3 \cdot \cos\left(x\sqrt{\frac{3}{4}}\right) + C_4 \cdot \sin\left(x\sqrt{\frac{3}{4}}\right) \right)$$

$$= C_1 \cdot \cos\left(\frac{x\sqrt{3}}{2}\right) + C_2 \cdot \sin\left(\frac{x\sqrt{3}}{2}\right)$$

$$+ x \left(C_3 \cdot \cos\left(\frac{x\sqrt{3}}{2}\right) + C_4 \cdot \sin\left(\frac{x\sqrt{3}}{2}\right) \right)$$

e.g. Solve $y^{(4)} + y''' + y'' = 0$

Solⁿ: Let $y = e^{mx}$ $\therefore y' = m e^{mx}$ and $y'' = m^2 e^{mx}$

Substituting these into the DE:

$$m^4 + m^3 + m^2 = 0$$

$$\Rightarrow m^2 (m^2 + m + 1) = 0$$

$$\Rightarrow m=0 \quad | \quad \Rightarrow m=0$$

Roots repeat

$$\Rightarrow m^2 + m + 1 = 0$$

$$\Rightarrow m = \frac{-1 \pm \sqrt{(1)^2 - 4 \cdot 1 \cdot 1}}{2 \cdot 1}$$

$$= \frac{-1 \pm \sqrt{-3}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$$

$$= \frac{-1}{2} \pm i \frac{\sqrt{3}}{2} \quad \leftarrow \text{Roots are imaginary}$$

$\therefore$ General solution:

$$y = A e^{0x} + B x \cdot e^{0x} + e^{\frac{-x}{2}} \cdot \left(C \cdot \cos\left(x \cdot \frac{\sqrt{3}}{2}\right) + D \cdot \sin\left(x \cdot \frac{\sqrt{3}}{2}\right) \right)$$
$$= A + Bx + e^{\frac{-x}{2}} \cdot \left(C \cdot \cos\left(x \cdot \frac{\sqrt{3}}{2}\right) + D \cdot \sin\left(x \cdot \frac{\sqrt{3}}{2}\right) \right) //$$